

Goals

- ▶ Introduce some ways of measuring effects
- ▶ Distinguish between measures of effect and measures of clarity
- ▶ Discuss advantages and disadvantages of standardized (unitless) measures of effect

Outline

Effects vs clarity

Absolute effects

Standardized effects

Scaling *predictors*

Conclusions

Model summary

An analogy

Standard deviation and standard error

A story

- ▶ A very cool biology seminar speaker told us this
 - ▶ Vitellogenin is a female hormone, male fish shouldn't have any
 - ▶ But we found that males in the disturbed site had *significantly more* vitellogenin
 - ▶ The picture showed about $4\times$ increase, but speaker did not emphasize this
- ▶ * Significance is a measure of clarity, not effect
- ▶ * $4\times$ is a measure of effect, but is it one that we understand?

A modest proposal

- ▶ One P value is sometimes useful as a summary of a substantive part of an analysis
 - ▶ We should never be looking at a lot of P values
- ▶ What could we do instead?
 - ▶ * Distribution of effect sizes
 - ▶ * Distribution of minimal effect sizes (based on confidence intervals)

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Summarizing effects

- ▶ In most cases, statistical analysis is focused on a one-parameter effect
 - ▶ Difference Or interaction
 - ▶ Slope
- ▶ Other cases are more complicated
 - ▶ Species distribution
 - ▶ Gene expression
 - ▶ Life-history tradeoffs

Simple effects

▶ Difference

- ▶ Mice in the treatment group gained on average 25mg (-2mg, 41mg) more weight than mice in the control group
- ▶ Male fish in the disturbed area had 7.2 ng/ml more vitellogenin than those in the undisturbed area
- ▶ Interaction: Scores in intervention group improved by 2.3 points (1.1pts, 5.0 pts) more than they improved in the control group

▶ Slope

- ▶ We estimated an increase of 4000 annual deaths per fluA percent positivity (2700d/fluApos, 5500d/fluApos)
- ▶ Soybean yield is estimated to increase 50kg/ha/°C in this temperature range (I'm tired of the CIs now, you get the idea)

Units ...

- ▶ are critical to interpreting these effect sizes
- ▶ Advantages
 - ▶ You need to engage with and understand these units
 - ▶ This drives biological understanding
 - ▶ And can force thoughtful choices about biological importance
- ▶ Disadvantages
 - ▶ You need to engage with and understand these units
 - ▶ Can be difficult, not always relevant
 - ▶ You can't compare across different systems
 - ▶ Are mice more dimorphic than elephants?
 - ▶ Which natural characteristics are under strong directional selection?

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Making unitless variables

- ▶ Divide by something with the same units
 - ▶ Mean, standard deviation, difference
- ▶ Take logs
 - ▶ * You're still dividing by something with the same units
 - ▶ * Once you log it, it doesn't matter what you chose

Example: correlation

- ▶ What is a correlation (ρ)?
 - ▶ * It is the slope of the relationship between two variables
 - ▶ * After both are divided by their sd
- ▶ What is R^2 ?
 - ▶ * it is literally ρ^2 – terrible nomenclature
 - ▶ * Reflects variation explained and ties to deep stuff about linear modeling

Proportional change

- ▶ If we divide a physical quantity (like tail length) by a group mean, we're now talking about proportional change
- ▶ Logs are a somewhat different (and generally more consistent) way to think about proportional change

Proportional change

- ▶ Proportional change is a bit weird:
 - ▶ If the females are 40% lighter than the males:
 - ▶ * the males are 66.7% heavier than the females
- ▶ Using (natural) logged ratios is a very appealing substitute
 - ▶ $\log(1.67)=0.511$; $\log(0.6)=-0.511$
 - ▶ JD would say this is a proportional change of 51%
 - ▶ But you might not want to try that

Digression: logged ratios

Mix and match

- ▶ Phenotypic evolution studies the effect of traits on measures of fitness
- ▶ The field decided to construct standardized slopes by
 - ▶ Dividing trait measures by their sd
 - ▶ Dividing fitness measures by their mean
- ▶ Why not?

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Scaling *predictors*

- ▶ Sort of split the difference between absolute and standardized
- ▶ If we have many predictors for one outcome, it may make sense to keep the units on the outcome only
- ▶ If we had one predictor and many outcomes, it might make sense to do the opposite
 - ▶ (e.g., CO₂)

Z scores

- ▶ Subtracting the mean and dividing by the standard deviation is the “standard” way of standardizing
 - ▶ You can also skip the subtraction step
- ▶ This allows comparison of the (estimated) effects of different predictors across the study
 - ▶ Related to their importance in predicting changes in outcome

Mean standardization

- ▶ If you instead divide by the mean you are looking at effects on the scale of the average amount of the predictor
 - ▶ Comparing to the counter-factual than an effect is absent

Influenza study

- ▶ In the influenza study, we mean-standardized our predictors but kept our outcome as deaths/year
 - ▶ * Effect size was deaths per year
 - ▶ * In other words, compared to the counter-factual of no influenza
- ▶ For weather variables, it might have made sense to use z-scores instead
 - ▶ * How much of the variation we saw in Chicago was explained by changes in winter temperatures
 - ▶ * As opposed to how much less death would we expect if there were no cold days in Chicago winters

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- ▶ Differentiate measures of effect from measures of clarity
 - ▶ Prefer the former when discussing scientific implications of your result
- ▶ Think carefully about whether to standardize
 - ▶ Pay attention to units if you keep them
- ▶ Think carefully about how to standardize
 - ▶ Make sure your interpretations are consistent with your choices